Exercise 1.2: Show that the inner product defined by Eq. 1.2 satisfies all the axioms of inner products.

Using the inner product, we can define some concepts that are familiar from ordinary 3-vectors:

- *Normalized Vector*: A vector is said to be normalized if its inner product with itself is 1. Normalized vectors satisfy,

$$\langle A|A\rangle = 1.$$

  For ordinary 3-vectors, the term *normalized vector* is usually replaced by *unit vector*, that is, a vector of unit length.

- *Orthogonal Vectors*: Two vectors are said to be orthogonal if their inner product is zero. |*A*> and |*B*> are orthogonal if

$$\langle B|A\rangle = 0.$$

  This is the analog of saying that two 3-vectors are orthogonal if their dot product is zero.

1.9.5 Orthonormal Bases

When working with ordinary 3-vectors, it is extremely useful to introduce a set of three mutually orthogonal unit vectors and use them as a basis to construct any vector. A simple example would be the unit 3-vectors that point along the *x*, *y*, and *z* axes. They are usually called $\hat{i}$, $\hat{j}$, and $\hat{k}$. Each is of unit length and orthogonal to the others. If you tried to find a fourth vector orthogonal to these three, there wouldn't be any—not in three